Advanced Computer Networks

Dr Sudipta Saha
Associate Professor
Dept of Computer Science & Engineering
Indian Institute of Technology Bhubaneswar

Queuing Theory: Examples

DSSRG: Decentralized Smart Systems Research Group

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Question – 1

1. Give the **rate transition diagram** for the **M/M/2** (i.e., 2 servers with infinite queue size) queueing system
   MARKS 10

2. Give the set of **equilibrium equations** for the **M/M/2** queueing system **MARKS 10**

3. Give an expression the **steady state probabilities p_1, p_k** for **$k \geq 2$ in terms of p_0** **MARKS 10**

4. Give an **expression for p_0 in terms of** the **arrival rate λ** and **service rate μ** (with a sum of infinite number of terms - work out the summation !!!) **MARKS 10**

5. Give an **expression** for the **average queue length in terms of** the **arrival rate λ** and **service rate μ** (with a sum of infinite number of terms - work out the summation !!!) **MARKS 10**

Q1.1

Give the **rate transition diagram** for the **M/M/2** (i.e., 2 servers with infinite queue size) queueing system

MARKS 10

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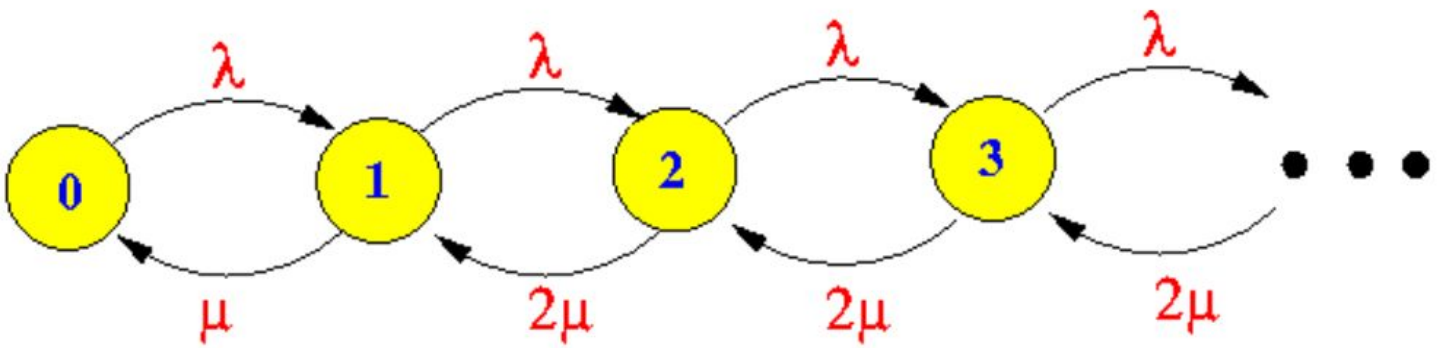
MARKS 10



Q1.2

Give the set of **equilibrium equations** for the **M/M/2** queueing system **MARKS 10**

Q1.2

Give the set of **equilibrium equations** for the **M/M/2** queueing system **MARKS 10**

- $p_0 \times \lambda = p_1 \times \mu$
- $p_1 \times \lambda = p_2 \times 2\mu$
- $p_2 \times \lambda = p_3 \times 2\mu$
- $p_3 \times \lambda = p_4 \times 2\mu$

...

Q1.3

Give an expression the **steady state probabilities p_1, p_k** for **$k \geq 2$ in terms of p_0** **MARKS 10**

Q1.3 Give an expression the **steady state probabilities** p_1, p_k for $k \geq 2$ **in terms of** p_0

MARKS 10

$$P_0 * \lambda = P_1 * \mu \quad => \quad P_1 = (\lambda/\mu) * P_0$$

$$P_1 * \lambda = P_2 * 2\mu \quad => \quad P_2 = (\lambda/2\mu) * P_1$$

$$= (\lambda/2\mu) * (\lambda/\mu) * P_0$$

$$= (\lambda^2/2\mu^2) * P_0$$

$$P_2 * \lambda = P_3 * 2\mu \quad => \quad P_3 = (\lambda/2\mu) * P_2$$

$$= (\lambda/2\mu) * \lambda^2/2\mu^2 * P_0$$

$$= (\lambda^3/4\mu^3) * P_0$$

Q1.3 Give an expression of the **steady state probabilities p_1, p_k** for **$k \geq 2$ in terms of p_0** **MARKS 10**

$P_1 = \lambda/\mu * P_0$

$P_2 = \lambda^2/2\mu^2 * P_0$

$P_3 = \lambda^3/4\mu^3 * P_0$

.

.

.

By observing the pattern, we can infer that…

$$\mathbf{P_k = (1/2)^{k-1} * (\lambda/\mu)^k * P_0 \quad = 2(\lambda/2\mu)^k * P_0}$$

Q1.4

Give an **expression for p_0 in terms of** the **arrival rate λ** and **service rate μ** (with a sum of infinite number of terms - work out the summation !!!)

Q1.4

P0 + P1 + P2 + P3 + … = 1

$P_0 + \lambda/\mu * P_0 + ½ * (\lambda/\mu)^2 * P_0 + ¼ * (\lambda/\mu)^3 * P_0 + \ldots = 1$

$P_0 * (1 + \lambda/\mu + ½ * (\lambda/\mu)^2 + ¼ * (\lambda/\mu)^3 + \ldots) = 1$

Consider $(\lambda/\mu) = x$

$P_0 * (1 + x + x^2/2 + x^3/4 + \ldots) = 1$

Consider this as S

$$S = 1 + x + x^2/2 + x^3/4 + \ldots$$

$$x*S/2 = x/2 + x^2/2 + x^3/4 + x^4/8 \ldots$$

$$S - (x*S/2) = 1 + x/2$$

Q1.4

$S - (x*S/2) = 1 + x/2$

$S * (1 - (x/2)) = 1 + x/2$

$S = (1 + x/2)/(1 - x/2)$

Earlier, $P_0 * (S) = 1$

$\Rightarrow P_0 * ((1 + x/2)/(1 - x/2)) = 1$

$\Rightarrow P_0 = (1 - x/2)/(1 + x/2)$

Replacing $x = (\lambda/\mu)$, we get

$\Rightarrow P_0 = (1 - \lambda/2\mu)/(1 + \lambda/2\mu) = \mathbf{(2\mu - \lambda)/(2\mu + \lambda)}$

Q1.5

Give an **expression** for the **average queue length in terms of** the **arrival rate λ** and **service rate μ** (with a sum of infinite number of terms - work out the summation !!!) **MARKS 10**

Q1.5

Average queue length, $N = \sum_k k^* P_k$

$$N = 0 * P_0 + 1 * P_1 + 2 * P_2 + 3 * P_3 + \ldots$$

$$= 1 * P_1 + 2 * P_2 + 3 * P_3 + \ldots$$

$$= 1*(1/2^0)*(\lambda/\mu)^1*P_0 + 2*(1/2^1)*(\lambda/\mu)^2*P_0 + 3*(1/2^2)*(\lambda/\mu)^3*P_0$$

$$= P_0*((\lambda/\mu) + (2/2)*(\lambda/\mu)^2 + (3/4)*(\lambda/\mu)^3 + (4/8)*(\lambda/\mu)^4 + \ldots)$$

Q1.5

Consider $(\lambda/\mu) = x$

$$= P_0*(x + (2/2)*x^2 + (3/4)*x^3 + (4/8)*x^4 + \ldots)$$

Consider $S = x + (2/2)*x^2 + (3/4)*x^3 + (4/8)*x^4 + \ldots$

$$xS/2 = (1/2)*x^2 + (2/4)*x^3 + (3/8)*x^4 + (4/16)*x^5 + \ldots$$

$$S-(xS/2) = x + x^2/2 + x^3/4 + x^4/8 + x^5/16 + \ldots$$

Q1.5

$S-(xS/2) = x + x^2/2 + x^3/4 + x^4/8 + x^5/16 + \ldots$

(Sum of G.P. with infinite terms = $a/(1-r)$) = $x/(1-(x/2))$

=> $S(1-(x/2)) = x/(1-(x/2))$

=> $S = x/(1-(x/2))^2$

Earlier $P_0(S) = P_0(x/(1-(x/2))^2)$

Replacing $x=(\lambda/\mu)$ and $P_0 = (2\mu - \lambda)/(2\mu + \lambda)$,

we get… $= \mathbf{(\lambda/\mu)/(1-(\lambda^2/4\mu^2))}$

Q1.6

Give an **expression** for the **average delay in terms of** the **arrival rate λ** and **service rate μ** **MARKS 10**

Q1.6

Average Delay, $T = N/\lambda$

$= ((\lambda/\mu)/(1-(\lambda^2/4\mu^2)))/\lambda$

$= \mathbf{(1/\mu)/(1-(\lambda^2/4\mu^2))}$

Question 2: Repair shop

- An **office** operates 5 **high speed printers**.
- Each printer has a **exponentially distributed** operational lifetime with an **average life time of 1/λ**. This will result in a **Poisson process** with **rate λ**.
- Printer failures are independent; so the **printer "failure rate"** when **k printers** are operational is equal to **k × λ**
- The office has a **printer repair shop** with **one printer repair technician**. The technician can repair a printer in an **average repair time of 1/μ**. This will result in a **Poisson process** with **rate μ**.

- Give the **rate transition diagram** for this Birth-death process (10 pts)

- Give the set of **equilibrium equations** (10 pts)

- Solve the **equilibrium equations** (**numerically**) and give the **steady state probabilities** (**numerical values**) for the case when **λ = 1** and **μ = 2** (10 pts)

- Compute (**numerically**) the **average queue length** for the case when **λ = 1** and **μ = 2** (10 pts)

- Compute (**numerically**) the **average arrival rate** of the **printers** to the **repair shop** for the case when **λ = 1** and **μ = 2** (10 pts)
- Average delay (repair time for a printer) in long term

2.1 Give the **rate transition diagram** for this Birth-death process (10 pts)

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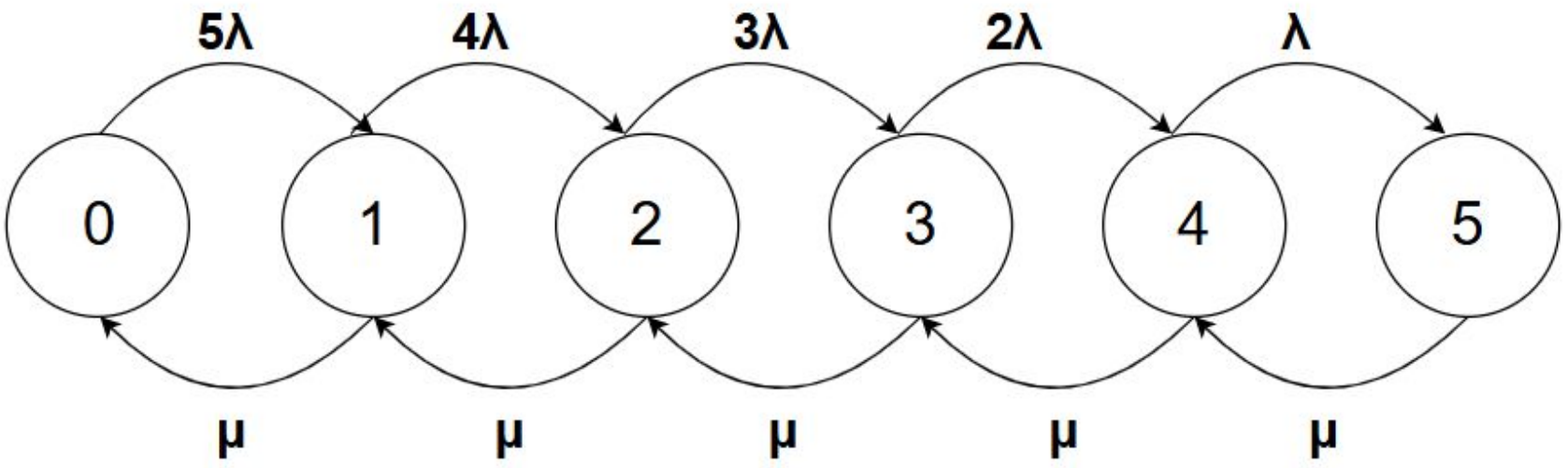
Solution:
Consider this scenario as a M/M/1/∞/N/FIFO Queue…
Where the printer repair technician acts as the server and the defective printers act as the packets/customers.

State (*n*): The number of broken printers in the system (waiting for repair + being repaired).
N = 5: Total number of printers.
λ: Failure rate per working printer.
μ: Repair rate of the technician.

2.1 Give the **rate transition diagram** for this Birth-death process (10 pts)

Solution:



2.2 Give the set of **equilibrium equations** (10 pts)

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- $p0 \times 5\lambda = p1 \times \mu$
- $p1 \times 4\lambda = p2 \times \mu$
- $p2 \times 3\lambda = p3 \times \mu$
- $p3 \times 2\lambda = p4 \times \mu$
- $p4 \times \lambda = p5 \times \mu$

2.3 Solve the **equilibrium equations** (**numerically**) and give the **steady state probabilities** (**numerical values**) for the case when **$\lambda = 1$** and **$\mu = 2$** (10 pts)

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- $P_0 \times 5\lambda = P_1 \times \mu$

=> $P_1 = (5\lambda/\mu) * P_0 = (5/2) * P_0$

- $P_1 \times 4\lambda = P_2 \times \mu$

=> $P_2 = (4\lambda/\mu) * P_1 = (20\lambda^2/\mu^2) * P_0 = (20/4) * P_0$

- $P_2 \times 3\lambda = P_3 \times \mu$

=> $P_3 = (3\lambda/\mu) * P_2 = (60\lambda^3/\mu^3) * P_0 = (60/8) * P_0$

- $P_3 \times 2\lambda = P_4 \times \mu$

=> $P_4 = (2\lambda/\mu) * P_3 = (120\lambda^4/\mu^4) * P_0 = (120/16) * P_0$

- $P_4 \times \lambda = P_5 \times \mu$

=> $P_5 = (\lambda/\mu) * P_4 = (120\lambda^5/\mu^5) * P_0 = (120/32) * P_0$

2.3 Solve the **equilibrium equations** (**numerically**) and give the **steady state probabilities** (**numerical values**) for the case when **$\lambda = 1$** and **$\mu = 2$** (10 pts)

- $P_0 + P_1 + P_2 + P_3 + P_4 + P_5 = 1$

=> $P_0 + (5/2) * P_0 + (20/4) * P_0 + (60/8) * P_0 + (120/16) * P_0 + (120/32) * P_0 = 1$

=> $P_0 = 1/27.25 = 0.036697$

2.3 Solve the **equilibrium equations** (**numerically**) and give the **steady state probabilities** (**numerical values**) for the case when **$\lambda = 1$** and **$\mu = 2$** (10 pts)

We got $P_0 = 0.036697$

Now, we can calculate…

$P_1 = (5/2) * P_0 = (5/2) * 0.036697 = 0.092$

$P_2 = (20/4) * P_0 = (20/4) * 0.036697 = 0.183$

$P_3 = (60/8) * P_0 = (60/8) * 0.036697 = 0.275$

$P_4 = (120/16) * P_0 = (120/16) * 0.036697 = 0.275$

$P_5 = (120/32) * P_0 = (120/32) * 0.036697 = 0.138$

2.4 Compute (**numerically**) the **average queue length** for the case when **λ = 1** and **μ = 2** (10 pts)

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Avg. queue length $L = (0) * P_0 + (1) * P_1 + (2) * P_2 + (3) * P_3 + (4) * P_4 + (5) * P_5$

$= 0 + (1)*(0.092) + (2)*(0.183) + (3)*(0.275) + (4)*(0.275) + (5)*(0.138)$

= 3.073 Printers

2.5 Compute (**numerically**) the **average arrival rate** of the **printers** to the **repair shop** for the case when **λ = 1** and **μ = 2** (10 pts)

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The arrival rate is state-dependent because broken printers cannot break again.

The arrival rate in state n is $\lambda_n = (5-n)*\lambda$.

Average Arrival Rate = $\sum((5-n)*\lambda) * P_n$

$= 5*P_0 + 4*P_1 + 3*P_2 + 2*P_3 + 1*P_4 + 0*P_5$

=1.93 (Printers/Unit Time)

2.6 Compute (**numerically**) the **average delay** of the **printers** in the **repair shop** for the case when **λ = 1** and **μ = 2** (10 pts)

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Average Delay = Avg. Queue Length/ Avg. Arrival Rate

= 3.073/1.93

= 1.59 unit time